Firewall Update: Tables for Two – Detailed Report

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Objective

The purpose of this challenge was to migrate the existing Linux firewall configurations from iptables to nftables on two virtual machines: Prod-Joomla and Fileshare. The goal was to review the existing firewall rules, recreate them using nftables, uninstall iptables, and make sure the new firewall configurations continued filtering traffic correctly. I also needed to allow incoming HTTPS traffic on TCP port 443 for Prod-Joomla and enable the nftables service so the firewall rules would remain active after a reboot.

Tools and utilities

- Linux Terminal
- GNU Nano
- nftables (nft)
- iptables
- apt package manager
- dpkg
- systemctl
- cat and ls
- pwd
- sudo

Task 1 – Prod-Joomla Linux Server

1. I logged into the Prod-Joomla Linux server and ran pwd to check my current directory. I also used ls to locate the existing firewall backup file.
2. I found the file in the home directory. I initially entered the filename incorrectly and received a "No such file or directory" error. After checking the filename again, I corrected the capitalization.
3. I ran cat ProdWeb_iptables_v4 to view the existing iptables configuration. The firewall had a default INPUT policy of DROP and allowed established and related connections, SSH on TCP port 22, and HTTP on TCP port 80. The 'forward' and 'output' policies were set to accept.
4. I reviewed the challenge requirements and confirmed that I needed to preserve the existing rules while also allowing incoming HTTPS traffic on TCP port 443.
5. I installed nftables using sudo apt install.
6. I edited /etc/nftables.conf using GNU Nano and configured the new firewall rules. I created an inet filter table with input, forward, and output chains. I set the input policy to 'Drop' and allowed established and related connections along with TCP ports 22, 80, and 443. I kept the forward and output policies set to 'accept'.
7. I initially tried validating the configuration using sudo nft -c -f /etc/nftables.conf but received an invalid option error because the installed version did not support the -c option.
8. I used sudo nft --help to review the available options.
9. I ran sudo nft list table inet filter to verify the firewall rules. The output confirmed that SSH, HTTP, and HTTPS were allowed and that the input chain had a default DROP policy.
10. I enabled the nftables service using sudo systemctl enable nftables. I initially entered the systemctl verification command incorrectly but corrected it to sudo systemctl is-enabled nftables and confirmed that the service was enabled.
11. I removed the old iptables package using sudo apt remove iptables. I then ran dpkg -l iptables and confirmed that the package was no longer installed.
12. I checked the nftables service using sudo systemctl status nftables and discovered that the service was inactive.
13. I started the service using sudo systemctl start nftables and checked its status again. The output showed active (exited) and confirmed that the configuration loaded successfully.

14. I checked the XP Cyber Range dashboard and noticed that the traffic-filtering check was still red even though the nftables rules were loaded and the service was enabled.
15. I rebooted the server to verify that the firewall configuration would remain active after restarting. After checking the challenge dashboard again, all three Prod-Joomla validation checks passed.

Task 2 – Fileshare Linux Server

1. I logged into the Fileshare Linux server running Ubuntu 24.04.3 LTS and ran pwd to check my current directory.
2. I used ls to locate the existing firewall backup file and found fs_iptables_v4 in the home directory.
3. I ran cat fs_iptables_v4 to review the existing iptables configuration. The firewall had a default input policy of drop and allowed established and related connections, SSH on TCP port 22, NetBIOS Session Service on TCP port 139, and SMB on TCP port 445. The forward and output policies were set to accept.
4. I ran sudo apt install nftables to make sure nftables was installed. The output confirmed that nftables version 1.0.9 was already installed.
5. I used ls /etc to locate the nftables configuration file and found nftables.conf.
6. I opened the configuration file using GNU Nano and recreated the existing firewall rules. I created an inet filter table with input, forward, and output chains.
7. I set the input chain policy to drop and added rules to allow established and related connections along with TCP ports 22, 139, and 445. I kept the forward and output policies set to ACCEPT to match the original configuration.
8. I saved the changes to /etc/nftables.conf and enabled the nftables service using sudo systemctl enable nftables. The terminal confirmed that the service was enabled by creating a symbolic link.
9. I removed the old iptables package using sudo apt remove iptables. The package manager confirmed that iptables was removed successfully.
10. I started the nftables service using sudo systemctl start nftables.
11. I ran sudo nft list ruleset to verify that the new firewall rules were loaded. The output confirmed that the input policy was DROP, established and related connections were allowed, and TCP ports 22, 139, and 445 were configured correctly.

12. I checked the XP Cyber Range dashboard and confirmed that all three Fileshare validation checks passed.

Final Results

During this challenge, I migrated the firewall configurations from iptables to nftables on two Linux virtual machines running different versions of Linux. I reviewed the existing firewall backup files, recreated the rules using nftables, and configured the input, forward, and output chains to preserve the original filtering behavior. I also added a new rule to allow HTTPS traffic on TCP port 443 for Prod-Joomla. I used GNU Nano to edit the configuration files, apt to remove the old iptables packages, systemctl to manage the nftables services, and nft commands to verify the firewall rules. I encountered a few problems along the way, including entering an incorrect filename, using an unsupported nft command option, and discovering that the Prod-Joomla nftables service was enabled but inactive. I worked through these issues by checking the available commands, correcting mistakes, starting the service, and rebooting the server to verify that the configuration persisted. After completing the migration, both servers passed all three of their XP Cyber Range validation checks, resulting in a Full Check Pass of 6/6**.** Overall, this challenge gave me much needed hands-on experience configuring Linux firewalls and migrating firewall rules as most of my previous labs used a graphic user interface when setting up firewall rules.